

## Statistics

Statistics is the language of numbers not just formulas

we need some  
tool which can  
summarise



data is doing

① mean  $\rightarrow \{210, 50, 60, 50, 70, 80, 70, \dots\}$  } major Qs / Ans

## Data types in statistics

① Continuous data: Data that can take any value within a range

0  $\longrightarrow$  1

0.1, 0.01, 0.001, 0.0001 (precision never ends)

example:

Value

Height

172.5 161.5 1500 152.52

② Discrete Data: Data that takes only countable, whole number values  
 $\rightarrow$  No decimals possible

example: no. of students (35), no. of exam questions (10),

③ Categorical data: Data that represents labels or categories

Nominal: data with no natural ranking between them

ex: Gender (Male / female / other)

Colors, Blood Group

Ordinal: data with a meaningful ranking or sequence

ex: Rank  $1^{st} > 2^{nd} > 3$

Medal: Gold, Silver, Bronze

④ Binary data: data that has exactly two possible values

ex.

0, 1

pass/fail

Yes/No

True/False

Spam/Not Spam

Fraud/Not fraud

## Types of Statistics

descriptive

It consists of organizing & summarizing the data

Inferential

Collect data  $\xrightarrow{\text{experiment}}$  Conclusion or inference

### ① Measure of Central Tendency

1. Mean

2. Median

3. Mode

### ② Measure of Dispersion

① Variance

② Standard Deviation

ex.

Sample Height

There is a college of 1000 students  
} 180 cm, 170 cm, 160 cm, 150 cm }

descriptive: mean, median, mode

avg height of a student

Inferential:

conclusion / inference

we conclude avg height of 1000 students after doing some experiments

## Measure of Central Tendency

① Mean (Average) : sum of all values divide total count

$N$  = population

$x$  = sample

$$\mu = \frac{\sum_{i=1}^N x_i}{N}$$

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

eg.

Ages =  $\{1, 3, 4, 5\} \rightarrow$  Distribution  $\rightarrow$  some outliers

$$(\text{mean}) \mu = \frac{1+3+4+5}{4} = \frac{13}{4} = 3.25,$$

② Median : The middle value of sorted dataset

Algorithm

(1) Sort the values in ascending order

(2) Count the total no. of values ( $N$ )

(3) ( $N = \text{odd}$ )  $\frac{N+1}{2} = \text{result (median)}$

(4) ( $N = \text{even}$ ) take 2 center values, calculate their average

$\{5, 1, 4, 3, 100\} \rightarrow$  (1)  $\{1, 3, 4, 5, 100\}$

(4)  $N = 5$

(3)  $\frac{N+1}{2} = \frac{5+1}{2} = 3$

(4) 4 is median

$\{1, 3, 4, 100\}$

$\{1, 3, 4, 5\} \rightarrow N = 4$

1 2 3 4

$$\frac{3+4}{2} = 3.5 \text{ median}$$

③ Mode : the most frequently occurring value in a dataset

$\{1, 2, 3, 7, 8\} = 8 \text{ (mode)}$

Class A, B, C, D

School sports [10 sports competitions]

③ ① 1 2

⑩ — friends — shopping — T-shirt  
shop keeper

$\{1, 2, 4, 5, 6, 2, 5, 4, 2, 5, 2\}$   
2 — 4  
4 — 2  
5 — 3

Mode ← {Red, Blue, Green, Red, Green, Red}

## Summary

| Measure | Best for                 | Sensitive to outliers | Data type             |
|---------|--------------------------|-----------------------|-----------------------|
| Mean    | numerical, no outliers   | Yes                   | continuous / Discrete |
| Median  | numerical, with outliers | No                    | Continuous / Discrete |
| Mode    | Most frequent value      | Rarely                | All types             |